

Using Generative Artificial Intelligence to Improve User Engagement in Content Marketing

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Core Domain:	Generative Artificial Intelligence (GenAI), LLM Synthesis, Grounded Prompting, Content Retrieval
HAT-RAG Mapping:	Grounded Evidence Generator (generator.py) & Automated Faithfulness Verification Engine

1. Executive Summary & Core Research Problem

This J.UCS research article investigates the impact of Generative AI (GenAI) and Large Language Models (LLMs) on text synthesis, semantic relevance, and grounded document generation. The authors compare ungrounded standalone LLM generation with context-conditioned semantic retrieval. The empirical findings demonstrate that conditioning generative synthesis on domain-specific retrieved context yields a **34.2% increase in user engagement** and reduces hallucinations by **58.6%**.

2. Key Technical Innovations

- Context-Conditioned Evidence Generation:** Combines structured document contexts with generative LLMs to produce accurate responses.
- Multi-Tier Content Abstraction:** Structures text output into multi-tier semantic summaries tailored to diverse user abstraction levels.
- Automated Grounding Metrics:** Establishes quantitative criteria for checking whether generated claims match source text facts.
- Hallucination Control:** Proves that structured evidence injection eliminates factual drifts in long-form generation.

3. Architectural Mapping to HAT-RAG CUDA Repository

J.UCS Concept	HAT-RAG-CUDA Module	Architectural Synergy & Impact
Evidence-Grounded Generation	src/generator.py Multi-Document Synthesis	Synthesizes final LLM responses using retrieved cross-document hierarchical contexts with explicit citation tags.
Multi-Tier Content Abstraction	src/hierarchical_tree.py Abstract Summary Hierarchy	Structures document corpora into Root Abstracts (Level 2), Local Abstracts (Level 1), and Leaf Passages (Level 0).
Hallucination Control	src/generator.py Faithfulness Score Engine	Calculates n-gram and term coverage between generated claims and retrieved source contexts to verify factual correctness.
User Engagement & Relevance	src/retriever.py MMR Reranking & Diversity	Filters redundant passages and selects diverse cross-document evidence to maximize answer completeness.

4. Benchmark Results & Research Synergy

Quantitative Benchmark: Grounded AI content achieved 34.2% higher engagement and 58.6% lower hallucination rates.

Synergy with HAT-RAG: Validates the primary objective of HAT-RAG-CUDA: combining multi-tiered hierarchical text retrieval with evidence-grounded LLM synthesis delivers vastly superior context quality, lower token consumption, and verifiable factual accuracy.